

Fernando Silva da Luz

Software Developer · Full-Stack & Gameplay

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PROFILE

Software engineer with 13+ years across full-stack, backend, computer vision and game development. I ship production systems in C#, Python and TypeScript — serverless backends (FastAPI / Azure), real-time computer-vision pipelines (Detectron2 / OpenCV / TensorFlow) and interactive 3D applications. B.Sc. Computer Science, 7× hackathon winner, and a Unity specialist who enjoys building games as much as the systems behind them.

CORE SKILLS

Languages: C#, C++, Python, TypeScript, JavaScript, GDScript, PHP, SQL

Backend & Cloud: FastAPI, Flask, Azure Functions, PlayFab, REST, SocketIO, Docker, Kubernetes, Azure, GCP, AWS

Data & Computer Vision: OpenCV, Detectron2, TensorFlow, MySQL, MongoDB

Game, 3D & Systems: Unity3D, C#, Godot, real-time 3D, AR / VR / MR, C++ / Qt desktop apps, CMake, Conan

Practice: Git, CI/CD, Agile · 7× hackathon winner

EXPERIENCE

Senior Software Developer

Aug 2023 – Present

Zimmer Biomet · Montréal, QC

- Build applications for robotic-assisted surgery — knee, hip, shoulder and brain — in C++, Qt (5/6), Python, CMake and Conan.
- Deliver performance-critical, regulated-quality software across the product lifecycle; started on the HoloLens 2 / Unity mixed-reality team, then moved into surgical robotics.

Senior Software / Unity Developer

Jun 2022 – Jul 2023

Prologue AI · Montréal, QC

- Built the serverless backend for CrimeTrip — FastAPI, Azure Functions and PlayFab for players and content.
- Designed modular, event-driven client architecture and real-time collaboration services.
- Optimized performance and cloud costs across the stack.

Software Engineer — Computer Vision

Aug 2020 – Dec 2021

BairesDev · Remote

- Built deep-learning pipelines for real-time detection, measurement and classification of physical objects for the real-estate market.
- Stack: Python, Detectron2, OpenCV, TensorFlow.

IT Director / Developer

Mar 2018 – Nov 2019

RSTcom Comunicação Estratégica · São Paulo, BR

- Led a development team; owned budgets, schedules and delivery for the events market.
- Shipped 12+ mobile apps to the App Store and Google Play.

Freelance Software Developer

Nov 2019 – Present

Freelance · Montréal / Remote

- Deliver full-stack and backend systems, APIs and interactive apps (Python, C#, TypeScript).

Earlier experience: *Founder @ Virtual Light · Senior Unity developer @ Expansão · Developer / test analyst @ TeleIDEA · Bilingual technical support @ Hewlett-Packard (19 enterprise clients).*

SELECTED GAMES & PROJECTS

CrimeTrip — Unity · AR · FastAPI · Azure · PlayFab

AR true-crime adventure game — investigate six cold cases inside augmented-reality crime scenes. prologuexr.com/crimetrip

Viva La Execution! — Unity · FastAPI · OpenAI · ElevenLabs

Comedic narrative game set in the French Revolution, with dynamic AI-driven dialogue. danyboyyy.itch.io/viva-la-execution

Down Below — Unity3D · C# · Turn-based combat

Turn-based JRPG inspired by Darkest Dungeon — party builder, combat and loot systems. vimeo.com/697498158

Janos — Unity · Oculus Rift · Game Jam

VR puzzle game built comfort-first to prevent motion sickness — a two-headed robot escape room. Made at SPJam. youtube.com/watch?v=eFgFCMwm-SU

Missão Fundo do Mar — Unity · Computer Vision · Video-mapping

Interactive projection wall using AI image recognition to bring children's drawings to life (Monica's Park). vimeo.com/359048835

HACKATHONS

1st place — EIT Digital DeepHack (Olivetti) — Trento, Italy (2019)
1st place — HackaEngage — Campus Party (2019)
1st place — FIESP Hackathon — Digital Government (2019)
1st place — VISA Hackathon — Agitech (2017)
2nd place — IBM BlueHack (2017)
2nd place — Synvia Hackathon (2019)
2nd place — Uber Hack (2019)

EDUCATION

B.Sc. Computer Science · Faculdades Metropolitanas Unidas (FMU)

2020 - 2023

B.A. Game Design · Universidade Anhembi Morumbi

2012 - 2016

CERTIFICATIONS

- Unity Certified Developer (2016) — among the first in the world
- Python Programming for SysAdmins — 4Linux (2019)
- Python Fundamentals — 4Linux (2019)

LANGUAGES

Portuguese — Native **English** — Fluent (bilingual) **French** — Intermediate (improving)